

A new method for observing the morphology of vesicular–arbuscular mycorrhizae

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A new procedure using chlorazol black E has been developed for staining vesicular–arbuscular mycorrhizal fungi in cleared roots. In a comparative study, chlorazol black E was found to be much superior to previously used stains (acid fuchsin, trypan blue, aniline blue) for showing details of internal hyphae and particularly arbuscules. This clearing and staining procedure, combined with Nomarski interference contrast microscopy, revealed details of arbuscule structure not evident with previous techniques. Field-collected samples were also stained well by this procedure. The procedure should allow more accurate assessment of roots for colonization by vesicular–arbuscular mycorrhizal fungi.

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Une nouvelle technique de coloration utilisant le noir de chlorazol E a été mise au point, afin de révéler à partir de racines éclaircies les entités fongiques appartenant aux endomycorhizes à vésicules et arbuscules. Lors d'une étude comparée le noir de chlorazol E confronté aux autres colorants d'usages (fuchsine acide, bleu de trypan, bleu d'aniline) s'est franchement distingué en montrant plus clairement les détails structuraux des arbuscules et des hyphes internes. Cette procédure d'éclaircir et de colorer les racines, jumelée avec la microscopie de contraste interférentielle (Nomarski) a permis de révéler avec plus de précision la morphologie des arbuscules par rapport aux techniques employées auparavant. Les échantillons provenant du champ et traités suivant la présente technique de coloration ont aussi donné de bons résultats. De plus, ce nouveau procédé devrait également faciliter les observations des racines hôtes au cours du processus de colonisation par les champignons endomycorhiziens.

Introduction

Vesicular–arbuscular mycorrhizae (VAM) are the most prevalent of the root–fungus symbioses (Smith 1980). The few genera of fungi involved form associations with the majority of vascular plants, including most crop and horticultural species. Since the fungal symbionts do not induce macroscopic changes in the root system, various microscopic techniques must be used to determine the presence and prevalence of these fungi (Carling and Brown 1982). The development of simple but reliable techniques is critical to the study of VAM since it is imperative to have a good understanding of the level and nature of root colonization in physiological, developmental, and pathogen–interaction studies. Also, since there are few diagnostic features of the mycosymbiont on which to base a classification scheme (Trappe and Schenck 1982), morphological features of fungal structures within the root may become important taxonomic characters.

The procedures used for rapid assessment of roots for VAM involve the clearing of roots with hot KOH followed by various staining procedures which selectively stain fungal structures (Bevege 1968; Phillips and Hayman 1970; Kormanik and McGraw 1982). In this paper we introduce the use of the stain, chlorazol black E, as a substitute for all previously used stains and show that it has wide applicability for a variety of host roots and several vesicular–arbuscular endophytes. Its use also allows the detection of all stages in the development of the symbiotic association with more clarity than has been possible using other staining procedures.

Materials and methods

Plant materials

Pot cultures of VAM

The VAM fungi *Glomus mosseae* (Nic. and Gerd.) Gerd. and Trappe, *Glomus versiforme* (Karst.) Berch, and *Glomus monosporum* Gerd. and Trappe were multiplied on leek (*Allium porrum* L.) roots. Leek seedlings were then inoculated with 1–2 g of roots from the

stock cultures in 20-cm plastic pots containing Turface (Plenchette *et al.* 1982). All leek pot cultures were grown in a growth room with a 16-h day length at $200 \mu\text{E m}^{-2} \text{s}^{-1}$ and a temperature of 21°C . Each pot was fertilized with 100 mL of Long Ashton solution weekly and watered with deionized water, following the procedure of Plenchette *et al.* (1982).

Field samples

Mature plants of wakerobin (*Trillium erectum* L.) and baneberry (*Actaea rubra* Ait. Willd) and white ash seedlings (*Fraxinus americana* L.) were harvested from a maple–beech forest near Waterloo, Ont. ($43^\circ 25' 30'' \text{N}$, $80^\circ 39' 30'' \text{W}$). Red clover (*Trifolium pratense* L.) plants from an adjacent hayfield were also sampled. The soil at this site is classified as a Fox sandy loam (Presant and Wicklund 1971).

Corn plants (*Zea mays* L. hybrid PAG SXIII) were grown in cores of field soil (Conestoga silty clay loam) in a growth chamber (O'Halloran 1982).

Stains

The following stains were used: chlorazol black E (direct black 38, Colour Index (CI) No. 30235), trypan blue (CI No. 23850), acid fuchsin (CI No. 42685), and aniline blue WS (cotton blue, CI No. 42755). Acid fuchsin and trypan blue were obtained from Allied Chemical Corporation, aniline blue WS from Fisher Scientific Company, and chlorazol black E from Allied Chemical Corporation and Sigma Chemical Company.

Preparation of endomycorrhizal roots for light microscopy

Chlorazol black E is introduced as a substitute for acid fuchsin or trypan blue in the VAM clearing and staining technique which follows. (i) Wash roots thoroughly and fix overnight, or store in Formalin – acetic acid – alcohol (FAA) (Sass 1958). (ii) Rinse with several changes of tap water to remove FAA. (iii) Transfer samples into a 10% (w/v) KOH solution in autoclave-resistant jars. Clear in autoclave for 15 min at 121°C . Samples with delicate roots may require shorter clearing times. (iv) Rinse with several changes of tap water followed by deionized water. (v) Transfer roots into a staining solution consisting of equal volumes of 80% lactic acid, glycerin, and distilled water with 0.1% (w/v) chlorazol black E. Stain for 1 h or longer at approximately 90°C . Prepare the staining solution several hours before use and allow undissolved stain particles to settle out. Do

not breathe dust from the stain.¹ The staining solution may be used several times before adding more stain if filtered through cheesecloth. A 60-min staining time is usually adequate, but longer staining will aid morphological studies. Samples previously stained with acid fuchsin and stored in glycerin may be restained with chlorazol black E. (vi) Transfer stained roots into glycerin for storage or observation with a transmitted light dissecting microscope. (vii) After destaining overnight in glycerin, mount roots on slides using Hoyer's mounting fluid with the chloral hydrate reduced to 2% (see Cunningham (1972) for recipe and instructions).

Evaluation of staining ability

To evaluate the staining ability of chlorazol black E, the following were examined: (i) the comparative staining of endomycorrhizal leek plants using a number of dyes; (ii) morphological details of several *Glomus* species associated with leek plants; and (iii) structural variations found in VAM from field samples of a number of vascular plant species.

Stain comparison experiment

For this experiment, a leek — *G. mosseae* pot culture was harvested 6 months after inoculation and the roots were divided into four uniform samples. Each sample was stained with one of the following dyes: acid fuchsin; aniline blue WS; trypan blue or chlorazol black E. All the dyes were used at 0.1% (w/v) in lactoglycerol, following the described staining procedure. After staining for 1 h the samples were allowed to destain in glycerin for 24 h, except for those stained with acid fuchsin, which were photographed after 1 h. A green filter was used to increase contrast when photographing the acid fuchsin stained material. Otherwise all of the samples were photographed with the same procedure and printed at the same contrast.

Morphology of VAM

Roots from 2- to 8-month-old leek pot cultures were cleared and stained following the described procedure. Morphological features and developmental stages of the endophytes were observed with emphasis on arbuscule structure.

VAM in field samples

Several naturally infected host plants were included to compare clearing and staining with leek plants grown in a growth room.

Photography

Cleared and stained roots mounted in Hoyer's medium were photographed using a Leitz Orthoplan photomicroscope. Thin mounts made from squashed roots or root fragments were photographed with a Zeiss photomicroscope III using Nomarski interference contrast microscopy.

Results

Stain comparison experiment

In Figs. 1 to 12, unless otherwise stated, the endophyte is *G. mosseae* and standardized staining and photography procedures were used. Staining of the internal hyphae and arbuscules was compared since all four stains resulted in intense staining of external hyphae and vesicles.

When acid fuchsin is used, fungus structures can be identified in the roots, but staining intensity is low (Figs. 1 and 2). Staining intensity fades rapidly in glycerin or Hoyer's mounting medium. Arbuscules could only be seen clearly within a few hours of staining and observation was complicated by excess stain diffusing out of the roots.

It was possible to obtain better results with acid fuchsin when a concentration of 0.5% in lactoglycerol was used and contrast was increased during the printing process. Figure 3 shows a

young infection unit of *G. monosporum* in leek, which is typical of the best results we have obtained with this stain.

Aniline blue WS (cotton blue) produced better results than acid fuchsin. The staining intensity did not fade rapidly but was still light (Figs. 4,5,6). Figure 6 shows that this stain does reveal some details of arbuscule structure.

Trypan blue (Figs. 7,8,9) produced better results than aniline blue WS or acid fuchsin. Staining of the fungus did not fade in glycerin or Hoyer's mounting medium. Arbuscule structure is revealed and there is no difficulty in following the distribution of the fungus in the host root.

Chlorazol black E (Figs. 10,11,12) produced the best results of the four stains used. The intense staining of internal hyphae and arbuscules facilitated observation of *G. mosseae* in leek roots and the black colour of the stain resulted in better contrast in photography. There was less background staining with chlorazol black E so that roots could be assessed for the presence of mycorrhizae immediately after staining.

Intensity of the four stains was compared 1 month after staining. Fading was not observed in roots stained with chlorazol black E, aniline blue WS, or trypan blue, when they were stored in glycerin or in Hoyer's medium. The roots stained with acid fuchsin had faded until only vesicles and external hyphae could be seen.

Morphology of VAM

Mycorrhizal leek plants grown under controlled conditions and colonized with known endophytes were used as a further test of the chlorazol black E staining procedure. Material from pot cultures was easy to clear because the leek roots were almost colourless and the surface washed off easily.

After staining with chlorazol black E it was possible to identify internal hyphae, arbuscules and vesicles at relatively low magnifications (Figs. 13,14). In early stages of the association many arbuscules and internal hyphae were observed (Fig. 13), while in older associations vesicles predominate (Fig. 14). Arbuscules were observed less frequently in older pot cultures (6–8 months) despite the higher levels of colonization.

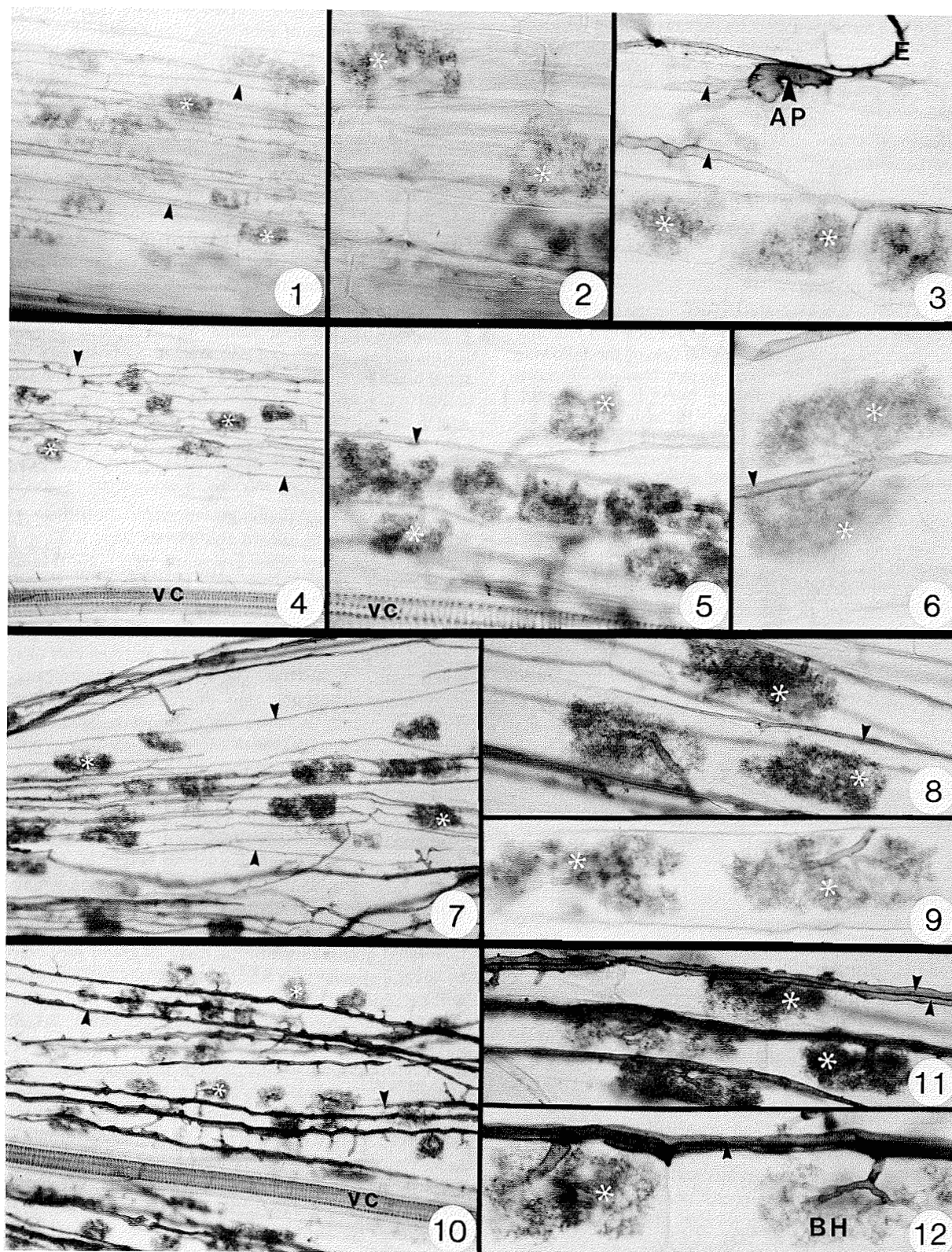
Arbuscular structure is observed in greater detail when Nomarski interference contrast microscopy is used in combination with chlorazol black E staining (Figs. 15,16,17,18).

The quality of staining is most critical when young internal hyphae and the earliest stages of arbuscule formation are being observed (Fig. 15). The use of chlorazol black E has greatly facilitated observations of the early stages of root colonization by several species of VAM fungi. Results of these observations will be reported elsewhere.

Arbuscule formation was observed with several endophytes and stages in the formation of *G. mosseae* arbuscules are representative of the results obtained (Figs. 16,17,18). Details of arbuscule structure are easier to observe in cells containing developing arbuscules (Figs. 16,17) than in cells containing mature arbuscules (Fig. 18). For example, it is possible to visualize the network of fine hyphae that comprise the arbuscule in Fig. 17. Mature arbuscules are harder to observe because the host cell is often occluded by arbuscule branches (Fig. 18).

This procedure also stains the lignified walls of xylem tracheary elements and cells with suberized walls such as the endodermis (Fig. 19). However, mycorrhizal fungi are not usually obscured by the vascular cylinder if the root is squashed under a cover slip.

¹Chlorazol black E along with acid fuchsin, trypan blue, and other biological stains are suspected of being carcinogenic (Combes and Haveland-Smith 1982). Because of variations in the dye content of chlorazol black E from different suppliers, changes in stain concentration and staining time may be required to obtain the best results.



VAM in field samples

Variations of endophyte morphology in different host species were observed when field-collected samples were stained with chlorazol black E. Difficulties were encountered because many of the naturally grown roots either were pigmented or contained an exodermis which stained with chlorazol black E. Autoclaving the roots for 15 min did not remove all of this natural colour, but further clearing damaged the arbuscules. The best results were obtained when thin mounts or root fragments were observed after clearing and staining in the usual manner.

The roots of wakerobin (*T. erectum*), a woodland plant, were heavily colonized by mycorrhizal fungi (Fig. 20). Numerous arbuscules, comprised of very fine branches, were present, but some cortical cells also contained coils of hyphae (Fig. 21).

The woodland plant banberry (*A. rubra*) had a brown epidermis that obscured internal details. In the material sampled, arbuscules were scarce and the majority of root cortical cells were filled with irregularly branched coarse hyphae. Figure 22 shows part of a squashed root in which cells containing coarse hyphae or arbuscules can be seen lying over the epidermis of the root.

White ash (*F. americana*) seedlings were heavily colonized, but the fungus structure was hard to observe because the roots contained a darkly staining exodermis. Several layers of thin, convoluted internal hyphae were often found between cortical cells, some of which contained arbuscules (Fig. 23).

Field-grown red clover (*T. pratense*) roots stained darkly with chlorazol black E because of the presence of many arbuscules. The cortical cells were packed with arbuscule branches so that in many cases they appeared as black rectangles (Fig. 24). Only in young arbuscules could individual arbuscular branches be observed (Fig. 25).

Corn plants (*Zea mays*) were colonized by VAM when grown in samples of field soil. Corn roots were easy to clear but had a thick vascular cylinder which obscured some details. Arbuscule structure was revealed when Nomarski interference contrast optics were used to observe young arbuscules (Fig. 26).

Discussion

The clearing and staining of plant roots to reveal endophytic fungi is central to the study of VAM (Carling and Brown 1982; Kormanik and McGraw 1982). Bevege (1968) and Phillips and Hayman (1970) introduced a procedure that used heated KOH to clear whole roots prior to staining with trypan blue in lacto-

phenol. KOH removes the cytoplasm from host root cells to allow observation of arbuscules and internal hyphae. Modifications to this procedure include the use of an autoclave to reduce clearing time (Bevege 1968; Kormanik *et al.* 1980) and the substitution of lactoglycerol for lactophenol (Kormanik *et al.* 1980). This latter modification which still produces good results was a major improvement in safety because of the toxicity of phenol.

Most researchers use acid fuchsin or trypan blue to stain VAM in cleared roots (Kormanik and McGraw 1982), although the use of aniline blue WS (cotton blue) has also been reported (Nicolson 1959). There is no information in the literature on which of these stains is the most effective or on the use of stains other than acid fuchsin with lactoglycerol (Kormanik and McGraw 1982).

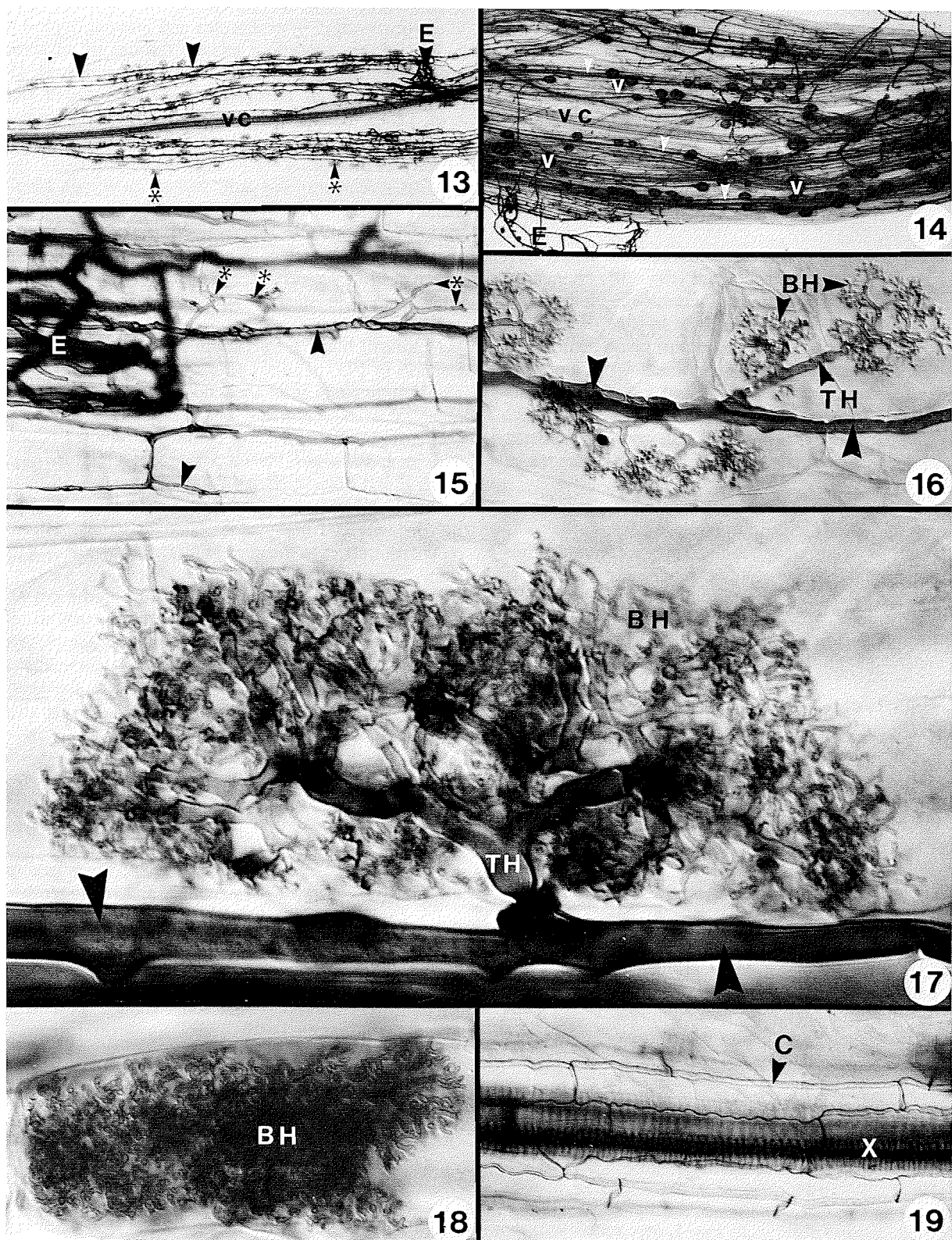
Chlorazol black E was introduced as a biological stain by Cannon in 1937 (Lillie 1977). Armitage (1943) found that it would stain fungi in lactophenol and was a valuable stain for plant material. Chlorazol black E has also been used as a chromosome stain in plant cytological studies (Conn 1943). Wilcox and Marsh (1964) reported that a mixture of chlorazol black E and Pianese III-B (malachite green, acid fuchsin, and martius yellow) was a good stain for paraffin sections of ectomycorrhizae and many other plant materials. There are few recent references to the use of chlorazol black E as a biological stain.

We decided to attempt staining of cleared roots with chlorazol black E because of its structural similarity to other dyes used for this purpose. All are amphoteric, anionic dyes (Gurr 1971). Chlorazol black E has a spectral curve that shows a uniform absorbance of light across most of the visible spectrum, while most other dyes, such as trypan blue, absorb light only in part of the visible spectrum (Gurr 1971, pp. 611 and 643). This characteristic results in higher contrast when chlorazol black E is used as a stain.

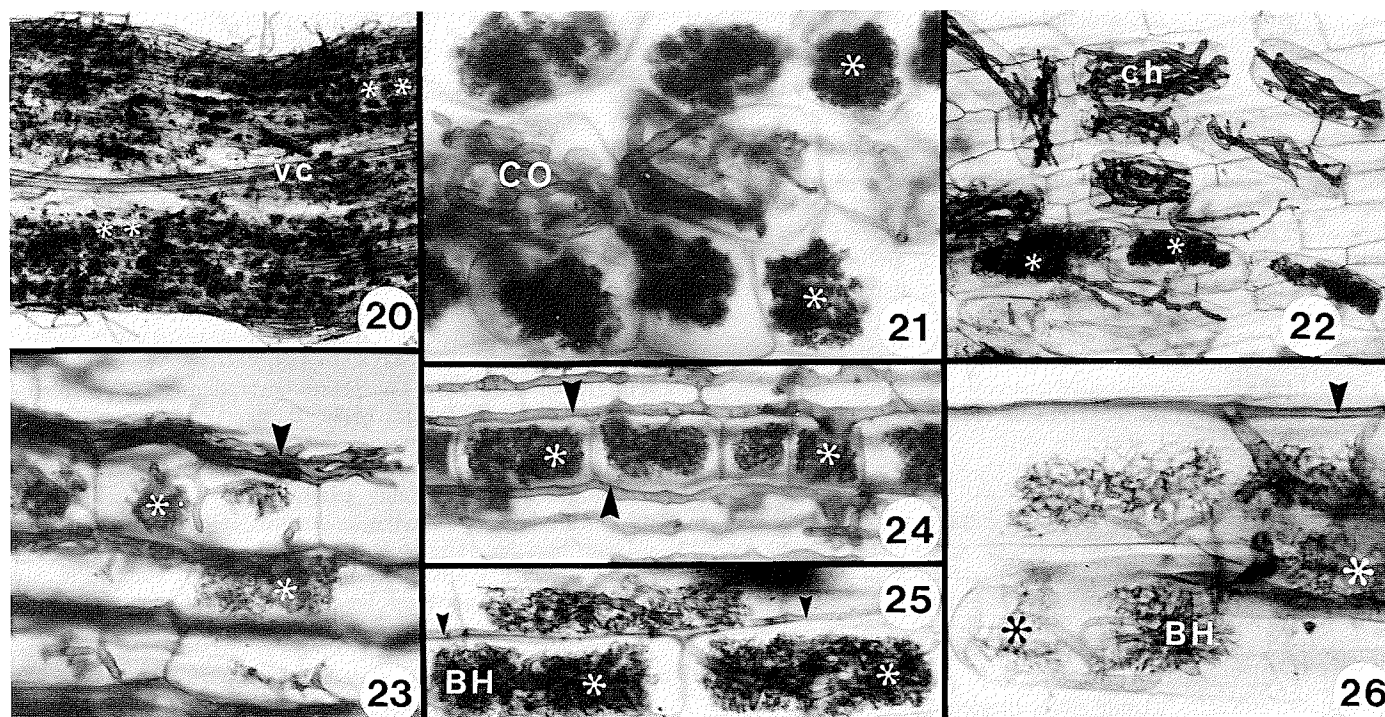
The improved staining of cleared VA mycorrhizal roots when chlorazol black E is used in place of acid fuchsin is probably due, in part, to a higher affinity of the former dye for fungal cell walls. Chlorazol black E has a linear structure (Gurr 1971) that may allow greater ionic and hydrogen bond interactions with fungal cell wall components. The rapid fading observed when material stained with acid fuchsin is stored in glycerin or Hoyer's mounting medium may be due to a lower affinity of this stain for fungal structures. Kormanik and McGraw (1982) report that roots stained with acid fuchsin could be stored in lactoglycerol for several years "with no

ABBREVIATIONS: *, arbuscule; arrowhead, internal hyphae; TH, trunk hyphae of an arbuscule; BH, branch hyphae of an arbuscule; vc, vascular cylinder; E, external hyphae.

FIGS. 1–12. Stain comparison experiment. Roots harvested from leek plants grown in growth room were cleared and stained with acid fuchsin, aniline blue WS, trypan blue, or chlorazol black E. All photographs, except Fig. 3, are of roots colonized by *Glomus mosseae*. Samples were stained and photographed with standardized procedures so that differences in the staining properties of the four stains could be observed. Figs. 1–3. Acid fuchsin stained roots. Fig. 1. Mycorrhizal root showing arbuscules and internal hyphae. $\times 100$. Fig. 2. Details of arbuscule structure are not very clear with this stain. $\times 250$. Fig. 3. Infection unit of *Glomus monosporum* composed of an appressorium (AP), internal hyphae, and arbuscules. $\times 250$. The material in Fig. 3 was stained with a higher concentration of acid fuchsin and printed at a higher contrast than the rest of the figures on this plate. Figs. 4–6. Aniline blue WS (cotton blue) stained roots. Fig. 4. Part of a colonized root showing arbuscules, internal hyphae, and the vascular cylinder of the root. $\times 100$. Fig. 5. Arbuscules and internal hyphae. $\times 250$. Fig. 6. Two arbuscules produced at one location on an internal hypha. $\times 400$. Figs. 7–9. Trypan blue staining. Fig. 7. Part of a flattened root. $\times 100$. Fig. 8. Higher magnification depicting arbuscules and internal hyphae. $\times 250$. Fig. 9. Trypan blue staining showing arbuscule structure in greater detail than acid fuchsin or aniline blue WS. $\times 400$. Figs. 10–12. Chlorazol black E staining of cleared roots. Fig. 10. Part of a mycorrhizal root showing internal hyphae, arbuscules, and the vascular cylinder. $\times 100$. Fig. 11. Arbuscules and internal hyphae are intensely stained by chlorazol black E. $\times 250$. Fig. 12. The fine branch hyphae of arbuscules are more apparent with this stain. $\times 400$.



FIGS. 13–19. Morphology of mycorrhizal leek roots obtained from plants grown in growth room, stained with chlorazol black E. Fig. 13. A young *Glomus mosseae* infection unit with many arbuscules. $\times 50$. Fig. 14. An old *G. versiforme* association composed of vesicles (v) and internal hyphae. $\times 50$. Figs. 15–19. Nomarski interference contrast microscopy. Fig. 15. A very young infection unit of *G. monosporum* with several developing arbuscules. $\times 220$. Figs. 16–18. Stages in the development of *G. mosseae* arbuscules. Fig. 16. Three young arbuscules showing trunk and some branch hyphae. $\times 340$. Fig. 17. An older arbuscule with numerous fine branch hyphae. $\times 1700$. Fig. 18. Host cell occluded by the branch hyphae of a mature arbuscule. $\times 910$. Fig. 19. The suberized casparian bands (C) of the endodermis and lignified walls of cells in the xylem (X) are also stained by chlorazol black E. $\times 220$.



FIGS. 20–26. Field-collected material cleared and stained with chlorazol black E. Figs. 20 and 21. Cleared and stained wakerobin (*Trillium erectum*) roots. Fig. 20. Root containing numerous arbuscules. $\times 30$. Fig. 21. Coils of hyphae (CO) and the densely staining arbuscules can be seen in cortical cells at a higher magnification. $\times 250$. Fig. 22. VAM of baneberry (*Actaea rubra*). Individual cortical cells, obtained by squashing cleared roots, filled with coarse hyphae (ch) or arbuscules. $\times 170$. Fig. 23. White ash (*Fraxinus americana*) seedlings. Several layers of convoluted internal hyphae are located between host cortical cells containing arbuscules. $\times 450$. Figs. 24 and 25. Field-grown red clover (*Trifolium pratense*). Fig. 24. Arbuscules of this fungus are composed of dense clumps of hyphae that completely fill the cortical cells. $\times 250$. Fig. 25. Younger arbuscules in which branch hyphae can be discerned. $\times 400$. Fig. 26. Corn roots grown in field soil. Branch hyphae of young arbuscules revealed by Nomarski optics. $\times 375$.

significant loss of stain in the fungus structures" which may indicate that low pH is necessary for maintaining the staining reaction.

The procedure used to clear roots also influences the final results. Clearing time must be adequate to remove all cytoplasm from host cells, but excessive clearing may damage delicate fungal structures. The use of an autoclave forces clearing agents into roots, resulting in more uniform clearing in less time.

Hoyer's mounting fluid is useful in VAM studies since it is miscible with water and glycerin and its refractive index is compatible with interference contrast microscopy. Cunningham (1972) reported that slides made with Hoyer's mounting medium will last for years. Mycorrhizal roots stained with chlorazol black E and mounted in this material show no signs of fading after 5 months. This allows slides of VAM to be stored for future reference.

Using the outlined clearing and staining procedure we have been able to observe the structure of VAM in considerable detail without using more complicated techniques. Since arbuscules can be observed easily in root cells of the host plant, this will allow more critical assessment procedures. For example, it should be possible to distinguish between developing, mature, and senescent arbuscules, for correlation with physiological data. This procedure is excellent for assessment techniques utilizing low magnifications achieved with a dissecting microscope (Giovannetti and Mosse 1980).

Many references on VAM state that they are prevalent in natural ecosystems; St. John and Coleman (1983), however,

found that there were few thorough scientific surveys to confirm this assumption. Detailed surveys of plants from diverse natural ecosystems, using clearing and staining procedures, are required. The limited field sampling included in the present study indicates the prevalence of VAM in the few species studied. The differences in VAM morphology observed in the field samples may have been caused by host root anatomy and (or) the endophyte species present.

The classification of VAM fungi currently relies in large part on anatomical features of soil-borne spores (Trappe and Schenck 1982). Abbott (1982) found, however, that it was possible to distinguish between 10 species of VAM fungi by comparing the anatomy of their infections in clover roots. It is probable that further detailed studies of the structure of VAM fungi in a variety of host roots will provide information that is of use in classifying this recalcitrant group of fungi.

Although many questions concerning VAM structure and development can be answered by light microscopy, a recent review (Carling and Brown 1982) emphasizes that most light microscopic observations have been confined either to confirming the presence of VAM associations or quantifying the association. There is obvious scope for research correlating the effects of different host–endophyte combinations and environmental conditions on VAM morphology (Carling and Brown 1982). In addition, detailed time course studies on VAM development, similar to those of Bevege and Bowen (1975), are required for a better understanding of this important symbiotic association. The chlorazol black E staining procedure should facilitate research in these areas.

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